

Modulation of Immune Response in Thermal Injury by Essential Fatty Acid-Deficient Diet

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This study was performed to determine the effects of essential fatty acid-deficient diet (EFAD) on immune parameters at 24 hours and 1 week after burn injury. Sprague Dawley rats were fed an EFAD diet or chow diet for 14 days. Twenty percent of the animals were put to death before the burn treatment (day 0), and the remainder were selected randomly to receive a scald or sham injury. Elevation in natural killer cell activity for EFAD-fed animals compared with rats fed the control diet was demonstrated at 100:1 effector:target (E:T) ratio on postburn day 1, and 25:1 E:T ratio on postburn day 7. EFAD-fed animals had significantly higher T-cell proliferation for all doses of phytohemagglutinin than did control animals before the burn injury. One week after the injury, EFAD-fed animals' T-cell activity was reduced to a level equivalent to that of control chow animals, whereas burned chow rats' lymphocyte activity was diminished significantly. EFAD diets enhanced T-cell proliferation and modestly improved natural killer cell response in both burned and control animals. (*J Burn Care Rehabil* 1996;17:465-70)

Immunologic compromise involving both cell-mediated and humoral immunity is well recognized in thermal injury.^{1,2} Trauma involves impaired neutrophil and natural killer (NK) cell function, depressed complement factors, and altered cytokine and prostaglandin production.³⁻⁶ Arachidonic acid metabolites have been implicated in postburn immunosuppression and may influence NK cell and cytokine activity.^{6,7}

NK cells are large, granular lymphocytes that are spontaneously cytotoxic against tumor cells, virus-infected cells, and some microbial pathogens. NK cells require no prior sensitization or restriction by the major histocompatibility complex before initiation of lytic activity.⁸ NK cells are able to synthesize IL-1, IL-2, and interferon- γ .⁹ In response to IL-1 activation, T-helper cells and NK cells produce IL-2, which primarily induces T-cell proliferation. NK cell proliferation and activity are primarily modulated by IL-2; however, other cytokines may be needed for

optimal costimulatory effects.¹⁰ NK activity may be enhanced after thermal injury by both IL-2 and interferon administration.¹¹

Dietary fat and metabolites of arachidonic acid have been shown to influence NK cell activity. In a study conducted by Leung and Ip,¹² NK cell activity was lower in rats fed a 20% corn oil diet for 10 days than it was in rats fed a 0.5% corn oil diet. Ingestion of eicosapentaenoic acid-ethyl ester has been shown to depress NK cell activity in rat spleen cells, and also lower the production of leukotriene B₄.¹³ NK activity is decreased in the presence of prostaglandin E₂ (PGE₂)^{7,14} and is enhanced by leukotriene B₄.¹⁵

Eicosanoid modulation by essential fatty acid-limiting states has been proposed to diminish the generation of proinflammatory lipid mediators. Manipulation of dietary polyunsaturated fatty acids has been used in several models of inflammation and shock.^{16,17} Essential fatty acid deficient diet (EFAD) rats have been shown to be more resistant to endotoxin challenge, with a significantly reduced mortality rate versus n-6 fatty acid-fed animals.¹⁷ Limiting essential fatty acid metabolism may be desirable for transient periods to minimize eicosanoid-related pathophysiologies. Therefore, this study evaluated the effects of dietary lipids on NK cell activity and lymphocyte blastogenesis among rats who were subjected to thermal injury.

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Table 1. Weight, food intake, and hematologic measures for day 0—animal and organ weights, food intake, and complete blood count by diet and treatment

Measure	CHOW	EFAD	p Value
Weights			
Arrival weight (gm)	243.8 ± 5.15	246 ± 5.15	0.72
Sacrifice weight (gm)	289.8 ± 5.35	286 ± 5.35	0.62
Food intake (gm/14 day)	265 ± 10.45	287 ± 10.45	0.16
Organ weights			
Spleen (gm)	0.65 ± 0.03	0.61 ± 0.03	0.40
Liver (gm)	11.78 ± 0.53	11.43 ± 0.53	0.65
Adrenal (gm)	0.06 ± 0.004	0.06 ± 0.004	0.84
Complete blood counts			
WBC ($\times 10^3/\mu\text{l}$)	4.74 ± 0.75	3.88 ± 0.75	0.43
Neutrophil (%)	14.4 ± 4.04	15.0 ± 4.04	0.92
Monocyte (%)	0.40 ± 0.40	0.40 ± 0.40	1.00
Lymphocyte (%)	85.2 ± 4.05	84.0 ± 4.05	0.83

METHODS

Diets and Animal Model. Sprague Dawley rats weighing between 240 and 255 grams were purchased from Sasco (Omaha, Neb.). Animals were housed in stainless steel hanging cages. Rats were kept in a temperature-controlled room with 12-hour alternating light and dark cycles. Water was provided ad libitum. Animals were assigned randomly to either the EFAD or the control diet (CHOW). Control animals were fed an AIN-76A rat chow diet, and EFAD animals were fed a modified AIN-76A fat-free formula obtained from ICN Biomedicals (Costa Mesa, Calif.). Individually housed animals were pair fed for 14 days, at which point 20% ($n = 16$) were put to death (day 0), and the remainder of the animals randomly were assigned burn ($n = 32$) or control ($n = 32$) treatments. The remainder of the animals were put to death on either postburn day 1 or 7. Animals were weighed on arrival and then every third day until their death. At the time of the animals' death, their organs were harvested (spleen, liver, and adrenal) and weighed. Spleens were placed in sterile petri dishes before initiation of the NK and lymphocyte proliferation assay. Blood samples were obtained from cardiac puncture. Blood smears were made with whole blood. The smears were fixed in methanol and stained with Diff Quik (Baxter Diagnostics Inc., McCaw Park, Ill.) for differential white cell counts. One hundred cells were counted per slide. At a

1:500 dilution, the number of white blood cells (WBC) was counted with a Coulter cell counter (Coulter Electronics, Hialeah, Fla.).

Model of Burn Injury. The burned rat model previously described by Walker and Mason¹⁸ was used in this experiment. Briefly, general anesthesia was achieved with sodium pentobarbital (0.64 mg/kg). The fur over the dorsum was clipped, and a full-thickness scald burn was produced on a template-exposed area by a 20-second immersion in 90° C water. A uniform burn of approximately 35% total body surface area was achieved. The animals were resuscitated with 10 ml of sterile lactated Ringer's injected subcutaneously. The control animals were treated in a similar fashion, except that they were immersed in tepid water.

NK Cell Assay. The NK assay was performed as previously described.¹⁹ Briefly, nonadherent splenic lymphocytes were isolated and used at effector:target (E:T) ratios of 100, 50, 25, and 12.5. YAC-1, a routine T-cell lymphoma (American Type Culture Collection, Rockville, Md.) was used as the target line of the assay and was maintained in log growth. Target cells were labeled with inorganic ⁵¹Cr ($\text{Na}_2^{51}\text{CrO}_4$). A constant 10^4 target cells were used in each culture well. Maximum ⁵¹Cr release was determined by adding 7.5% Triton-X detergent (Sigma, St. Louis, Mo.) to lyse all targets. Spontaneous ⁵¹Cr release was determined by adding culture media to target cells and counting radioactive label in the supernatant. Effector and target cells were incubated in a 5% CO₂ humidified chamber for 4 hours. Supernatants were collected by pipette and were counted for 1 minute on a gamma counter. The percentage of cytotoxicity was calculated with use of the following formula: % NK cytotoxicity = $100 \times \frac{\text{experimental release cpm} - \text{spontaneous release cpm}}{\text{maximum release cpm} - \text{spontaneous release cpm}}$.

Lymphocyte Proliferation Assay. A cell suspension was prepared from spleens that were teased apart under sterile conditions. Cells were washed twice in phosphate-buffered saline solution (PBS), counted, and adjusted to a final concentration of 5×10^6 cells/ml in fetal bovine serum.

One hundred μl of diluted cells were added in triplicates to a 96-well, flat-bottom sterile plate. Phytohemagglutinin (PHA) at various doses (0, 0.2, 2, and 20 $\mu\text{g}/\text{ml}$) was added to each well in triplicates. Plates were incubated for 48 hours at 37° C in a 5% CO₂ incubator. Plates were pulsed with 1 μCi of ³H per well for 24 hours. After 72 hours the cells were harvested on glass fiber filters. Each filter disk was placed in a scintillation vial. Five milliliters of scintil-

Table 2. Weight, food intake, and hematologic measures for burn day 1—animal and organ weights, food intake, and complete blood count by diet and treatment

Measure	CHOW		EFAD		p Value treatment	p Value diet	p Value treatment-diet
Postburn day 1	Control	Burn	Control	Burn			
Weights							
Arrival weight (gm)	240.6 ± 3.45	246.5 ± 3.45	243.3 ± 3.45	247.8 ± 3.45	0.15	0.56	0.84
Sacrifice weight (gm)	284.8 ± 4.18	290.6 ± 4.18	283.1 ± 4.18	287 ± 4.51	0.26	0.56	0.86
Food intake (gm/15 day)	284.7 ± 7.05	281.3 ± 7.06	287.3 ± 7.05	301.6 ± 7.05	0.15	0.56	0.84
Organ weights							
Spleen (gm)	0.62 ± 0.04	0.58 ± 0.04	0.71 ± 0.04	0.66 ± 0.04	0.15	0.31	0.53
Liver (gm)	12.12 ± 0.40	10.68 ± 0.40	11.78 ± 0.40	10.10 ± 0.44	0.002	0.26	0.79
Adrenals (gm)	0.06 ± 0.004	0.06 ± 0.004	0.06 ± 0.004	0.06 ± 0.004	0.19	0.60	0.89
Complete blood count							
WBC ($\times 10^{-3}/\mu\text{l}$)	3.29 ± 0.67	5.21 ± 0.67	4.12 ± 0.73	4.71 ± 0.67	0.09	0.83	0.35
Neutrophil (%)	9.40 ± 5.73	54.83 ± 5.23	12.66 ± 5.23	47.2 ± 5.73	0.0001	0.70	0.33
Monocyte (%)	1.40 ± 0.48	0.66 ± 0.44	0.66 ± 0.44	1.40 ± 0.48	1.00	1.00	0.13
Lymphocyte (%)	87.20 ± 5.87	44.50 ± 5.36	86.33 ± 5.36	51.40 ± 5.87	0.001	0.56	0.50

Table 3. Weight, food intake, and hematologic measures for burn day 7—animal and organ weights, food intake, and complete blood count by diet and treatment

Measure	CHOW		EFAD		p Value treatment	p Value diet	p Value treatment-diet
Postburn day 7	Control	Burn	Control	Burn			
Weights							
Arrival weight (gm)	249.6 ± 3.58	253.3 ± 3.58	249.5 ± 3.54	257.1 ± 3.58	0.12	0.62	0.60
Sacrifice weight (gm)	307.5 ± 8.66	308.5 ± 9.33	308.3 ± 8.66	33.3 ± 8.66	0.76	0.73	0.83
Food intake (gm/21 day)	404.3 ± 10.14	370.5 ± 10.92	395.2 ± 10.14	453 ± 10.14	0.25	0.002	0.0002
Organ weights							
Spleen (gm)	0.66 ± 0.04	0.71 ± 0.05	0.67 ± 0.04	0.83 ± 0.04	0.04	0.22	0.27
Liver (gm)	12.69 ± 0.42	12.34 ± 0.49	12.83 ± 0.42	13.36 ± 0.49	0.85	0.21	0.33
Adrenals (gm)	0.055 ± 0.004	0.057 ± 0.005	0.057 ± 0.004	0.060 ± 0.004	0.66	0.65	0.93
Blood counts							
WBC ($\times 10^{-3}/\mu\text{l}$)	3.51 ± 0.65	7.31 ± 0.76	4.16 ± 0.65	5.58 ± 0.76	0.001	0.44	0.09
Neutrophil (%)	14.51 ± 4.52	39.60 ± 5.35	15.33 ± 4.88	30.12 ± 4.22	0.0004	0.37	0.29
Monocyte (%)	1.71 ± 0.67	1.80 ± 0.79	0.33 ± 0.72	2.00 ± 0.62	0.23	0.41	0.28
Lymphocyte (%)	83.71 ± 4.30	58.60 ± 5.09	84.33 ± 4.65	67.87 ± 5.09	0.0001	0.29	0.35

lation fluid were added to each vial, and samples were counted on a scintillation counter.

Statistical Analysis. Repeated-measures analysis of variance was calculated for percentage of NK cytotoxicity across E:T ratios and lymphocyte blastogenesis over mitogen dose. Analysis of variance was used to determine whether dependent variables were influenced by diet or treatments (burn or control), or the interaction of diet and treatment (burn or control). Means were separated with the F-protected predicted difference test.²⁰ All statistical analyses were performed with general linear models procedures.²⁰

RESULTS

Weights and Food Intake. Initial weights and weights at time of death for the animals for day 0 and postburn days 1 and 7 were statistically similar among treatments (Tables 1, 2, and 3). Weight increased approximately 17% during the preburn feeding period for EFAD and CHOW animals. Food intake was not significantly different for EFAD and CHOW rats through days 0 or 1 (Tables 1 and 3). However, on day 7 the interaction between treatment and diet ($p = 0.0002$) was significant (Table 3). One week after injury, EFAD and CHOW rats had

similar food intake. Burn treatment affected food intake. On day 7, EFAD burned animals had significantly higher intakes compared with other diet and treatment groups. Organ weights did not differ on day 0. Burn treatment reduced liver weights compared with controls ($p = 0.002$) for postburn day 1 (Table 2). Burned animals put to death on day 7 had higher spleen weights compared with control animals ($p = 0.04$) (Table 3). All other organ weights were similar among groups.

Hematologic Measures. WBC count and differentials were statistically similar for day 0 for each diet and treatment group (Table 1). Burned animals' WBC counts were slightly elevated on postburn day 1. Burned rats had elevated WBC numbers by day 7 ($p < 0.001$). Neutrophils were elevated, and lymphocyte percentages were depressed for burned animals at 24 hours and at 7 days after burn injury (Tables 1 and 3).

NK Activity. Overall, diets had no significant effect on NK cytotoxicities. Before the burn injury, EFAD-fed rats had higher NK cytotoxicity, but this effect only approached significance with an E:T ratio of 12.5:1 ($p = 0.07$, Table 4). On postburn day 1, the EFAD diet enhanced NK activity at 100:1 E:T ratio ($p = 0.04$). On postburn day 7, animals on the EFAD diet had higher NK cytotoxicity ($p = 0.03$) at 25:1 E:T ratio, and approached significance ($p = 0.08$) at 12.5:1 E:T ratio. Burn injury suppressed NK activity at 50:1 and approached significance at 25:1 E:T ratio on day 7 (Table 5). The interaction between stress treatment was significant only for day 7, with an E:T ratio of 100:1 (Table 5).

Lymphocyte Transformation Assay. EFAD-fed animals had significantly higher blastogenesis activity for the average of all doses of PHA ($p = 0.02$) than did control animals for day 0 (Table 4). Burned rats had lower lymphocyte transformation assay ($p = 0.02$), compared with control animals, for postburn day 1 (Table 6). There were no significant diet or interaction effects (diet by treatment) for day 1. On day 7, a treatment effect was observed ($p = 0.04$) with reduced blastogenesis among burned animals. EFAD enhancement of lymphocyte activity was produced for a PHA dose of 20 $\mu\text{g}/\text{ml}$ ($p = 0.03$) and approached significance at 2 $\mu\text{g}/\text{ml}$ PHA ($p = 0.063$) at 7 days after burn injury (Table 5). The diet-treatment interaction was not significant, and burn treatment lowered—but EFAD enhanced—blastogenesis. Therefore the combined EFAD and burn group did not differ from the control and CHOW group.

Table 4. Immune parameters for day 0—NK cytotoxicity, lymphocyte transformation by diet and treatment

Measure	CHOW	EFAD	p Value
LTA: CPM			
All Dose PHA	15474 $\pm$ 6815	43178 $\pm$ 6815	0.02
0 $\mu\text{g}/\text{ml}$ PHA	15155 $\pm$ 13630	38735 $\pm$ 13630	0.25
0.2 $\mu\text{g}/\text{ml}$ PHA	16583 $\pm$ 13630	53075 $\pm$ 13630	0.07
2 $\mu\text{g}/\text{ml}$ PHA	13525 $\pm$ 13630	46857 $\pm$ 13630	0.11
20 $\mu\text{g}/\text{ml}$ PHA	16633 $\pm$ 13630	34046 $\pm$ 13630	0.38
NK % cytotoxicity E : T ratio			
All E : T ratio	9.82 $\pm$ 7.69	15.07 $\pm$ 7.69	0.67
12.5	4.76 $\pm$ 6.23	21.94 $\pm$ 6.23	0.07
25	5.52 $\pm$ 2.32	6.44 $\pm$ 2.32	0.79
50	13.17 $\pm$ 7.77	17.30 $\pm$ 7.77	0.75
100	14.56 $\pm$ 4.80	16.32 $\pm$ 4.80	0.80

LTA, Lymphocyte transformation assay; CPM, counts per minute.

DISCUSSION

Animal weights were similar on each day they were killed, in spite of a substantial burn treatment and pair feeding (Tables 1, 2, and 3). EFAD and EFAD-and-burned animals served as the control for food intakes. Food intakes were similar for days 0 and 1; however, treatment by diet interaction effect was seen on day 7 (Table 3). Burned EFAD animals' food intake determined food allotments for the other diet and treatment groups. EFAD burned animals ate significantly more compared to all other diet and treatment groups. EFAD-fed animals had a 14.6% weight gain after burn injury, whereas control burned animals lost 8.2% body weight. Mochizuki et al.,²¹ demonstrated similar results in burned guinea pigs fed 0% fat. Muscle mass and body weight at postburn day 10 were highest for animals on a fat-free enteral formula compared with diets with a 5% to 50% fat content. Two weeks after burn injury, nutritional indexes of serum albumin and transferrin were also the highest in 0% and 5% lipid groups, and the lowest in 30% and 50% lipid groups.²¹ Apparently the burn effect on EFAD-fed animals produced a hyperphagic response, which allowed for maintenance of body weight.

Organ weights did not differ for EFAD and control animals at day 0 (Table 1). The reduction in liver weights for burned animals (day 1) may be due to diminished perfusion of this organ in the shock phase of injury (Table 2). Elevated spleen weights in burned animals (day 7) may reflect an infectious process and the associated edema (Table 3). Pollock

Table 5. Immune measures for burn day 7 NK cytotoxicity and lymphocyte transformation by diet and treatment

Measure	CHOW		EFAD		p Value treatment	p Value diet	p Value treatment-diet
Postburn day 7	Control	Burn	Control	Burn			
LTA: CPM							
All Dose PHA	21276 ± 3855	8858 ± 4151	24430 ± 3855	19581 ± 3855	0.04	0.18	0.33
0 µg/ml PHA	27128 ± 5491	9949 ± 6340	28416 ± 5491	20850 ± 5491	0.04	0.30	0.41
0.2 µg/ml PHA	22957 ± 3795	9622 ± 4382	21149 ± 3795	16414 ± 3795	0.03	0.53	0.29
2 µg/ml PHA	23783 ± 5554	6158 ± 6413	28165 ± 5554	24260 ± 5554	0.07	0.06	0.24
20 µg/ml PHA	13038 ± 2925	9705 ± 3417	19991 ± 2959	16802 ± 2959	0.29	0.03	0.98
NK % cytotoxicity E : T Ratio							
All E : T Ratio	7.27 ± 1.81	5.30 ± 1.81	9.43 ± 1.43	7.78 ± 1.43	0.25	0.15	0.92
12.5	2.31 ± 1.63	0.80 ± 1.90	2.04 ± 1.63	6.64 ± 1.63	0.49	0.08	0.05
25	10.97 ± 3.16	3.49 ± 3.69	17.31 ± 3.16	12.70 ± 3.16	0.08	0.03	0.67
50	9.24 ± 2.26	1.15 ± 2.65	9.34 ± 2.27	6.74 ± 2.29	0.03	0.24	0.26
100	6.57 ± 2.34	13.18 ± 2.74	9.05 ± 2.34	5.03 ± 2.34	0.60	0.26	0.04

LTA, Lymphocyte transformation assay; CPM, counts per minute.

Table 6. Immune parameters for burn day 1—NK cytotoxicity and lymphocyte transformation by diet and treatment

Measure	CHOW		EFAD		p Value treatment	p Value diet	p Value treatment-diet
Postburn day 1	Control	Burn	Control	Burn			
LTA: CPM							
All Dose PHA	31148 ± 4336	21183 ± 4336	28691 ± 4336	17849 ± 4635	0.02	0.52	0.92
0 µg/ml PHA	32726 ± 4899	15613 ± 4899	32926 ± 4899	14172 ± 5237	0.001	0.91	0.87
0.2 µg/ml PHA	27852 ± 4488	15700 ± 4488	27195 ± 4488	16003 ± 4798	0.02	0.97	0.92
2 µg/ml PHA	34856 ± 5691	19253 ± 5691	33269 ± 5691	20824 ± 6084	0.02	1.00	0.79
20 µg/ml PHA	29156 ± 5575	34164 ± 5575	21376 ± 5575	20399 ± 5969	0.06	0.73	0.60
NK % cytotoxicity E : T ratio							
All E : T ratio	5.03 ± 2.07	3.09 ± 2.07	7.61 ± 1.93	8.19 ± 2.07	0.74	0.06	0.54
12.5	2.09 ± 1.19	2.48 ± 1.19	1.77 ± 1.19	3.31 ± 1.28	0.44	0.84	0.64
25	3.70 ± 3.20	4.84 ± 3.20	10.53 ± 3.20	3.84 ± 3.45	0.38	0.41	0.22
50	6.01 ± 2.52	3.31 ± 2.52	6.38 ± 2.52	7.17 ± 2.71	0.42	0.72	0.50
100	6.95 ± 4.36	6.77 ± 4.36	11.77 ± 4.36	21.04 ± 4.70	0.32	0.04	0.30

LTA, Lymphocyte transformation assay; CPM, counts per minute.

and Lotzova²² also found elevated spleen weights after surgical stress in mice. Adrenal weights did not differ significantly among treatments throughout the experiment.

Burned animals exhibited a classic pattern of inflammation in complete blood counts for days 1 and 7. Elevation in the percentage of neutrophils, with subsequent reduction in lymphocytes, was demonstrated after burn injury (Tables 2 and 3). WBC counts were slightly elevated 24 hours after the injury and were significantly different after 1 week (Tables 2 and 3). No diet or treatment effects were demonstrated before the burn injury (Table 1).

Manipulating dietary lipids alone had few effects on NK cytotoxicity. Consistently, EFAD-fed animals exhibited less NK suppression after the injury than did the controls, although this only approached significance ($p = 0.06$) (Table 6). On day 7, a diet effect was observed at an E:T ratio of 25:1 and approached significance at an E:T ratio of 12.5:1 (Table 5). Stress-induced suppression of NK activity was observed for animals 1 week after injury. Lowered NK activity may reflect the subsequent immunocompromise associated with modulated cytokine function and production. A stress and diet interaction effect was observed for an E:T ratio of 100:1,

and approached a significant effect at an E:T ratio of 12.5:1 on day 7. EFAD burned rats had improved NK cytotoxicity at the lowest E:T ratio; however, at an E:T ratio of 100:1, chow-fed burned rats had an elevation of NK activity. Inhibition of the cyclooxygenase pathway, and the subsequent reduction in PGE₂ synthesis, may modulate NK proliferation and activity. PGE₂ may inhibit lymphocyte production of IL-2 and therefore allow proliferation of suppressor T cells.^{5,23,24} Faist et al.,⁶ found that lymphocytes from thermally injured patients had decreased IL-2 secretion; however, when incubated with a cyclooxygenase inhibitor, IL-2 production was partially restored.

As previously reported by other investigators,^{25,26} EFAD-limiting states enhanced lymphocyte transformation compared with animals receiving essential fatty acids (Table 4). Stress of the burn injury depressed lymphocyte transformation on day 1 (Table 6). On day 7, lymphocyte transformation was elevated compared to day 1; however, burn suppression was still evident ($p = 0.04$). Lymphocyte transformation for EFAD-fed animals was elevated for 2 $\mu\text{g}/\text{ml}$ PHA ($p = 0.06$) and was significantly elevated for a PHA dose of 20 $\mu\text{g}/\text{ml}$ ($p = 0.03$) (Table 5).

During stress, T-cell activity was reduced among EFAD-fed animals to a level that was consistent with control nonburned animals. CHOW burned rats had depressed activity. Thus EFAD diets prevented the suppression of T-cell proliferation. Elimination of fat in this experimental model protected against the lowered NK cell activity and T-cell transformation reported in patient population groups with significant thermal injury.

In conclusion, stress reduced both T-cell proliferation and NK activity independent of diets. In addition, EFAD diets enhanced T-cell proliferation and improved NK response in both burned and control animals. The diminution of products from both the cyclooxygenase and lipoxygenase pathways may exert potential protective mechanisms during pathophysiologic states characterized by high eicosanoid production. Whether transitional EFAD diets are efficacious in thermal injury requires further study.

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